

Maria SALOP

PROFILE

Final-year Data Science student focused on building reliable, scalable AI solutions. Experienced across the full product lifecycle, from model integration and backend APIs to cloud deployment. Passionate about solving real-world challenges through applied AI and end-to-end engineering.

SKILLS

- Python (Pandas, Scikit-learn, PyTorch)
- SQL & Database Management
- Data Modelling & ETL Pipelines
- Machine Learning & Deep Learning
- Cloud Infrastructure (Alibaba Cloud, Azure)
- Containerization & Deployment (Docker)
- Statistical Analysis & Modeling
- Data Visualization (Power BI, Plotly)
- Mobile Development (React Native, Expo)
- Version Control (Git/GitHub)
- Agile Development Methodologies

CONTACT DETAILS

@ salopmaria37@gmail.com
☎ +31 621 613 890
🌐 linkedin.com/maria-salop
🐙 github.com/MariaSalop
📁 mariasalop.github.io
✉ Breda, North Brabant, Netherlands

PERSONAL INFORMATION

Citizenship: **Romania**
Languages: **English, Romanian**

EXPERIENCE

DATA DRIVEN CONFIGURATOR OPTIMIZATION INTERN

Vlisco

09.2026 – Present

◇ Analysing the product configurator that generates routings and bills of material for wax print fabric in an ERP environment. Auditing large table exports with Python and SQL to map the rule structure and identify redundant logic, validated against production history.

MACHINE LEARNING ENGINEER INTERN

Cognita Innovative Solutions

02.2026 – 06.2026

◇ Built and deployed AI components for a multi-product startup. Designed an OCR-based business card scanning pipeline (ML Kit) in React Native, implemented vector similarity search with Milvus on Alibaba Cloud, and integrated on-device Whisper speech-to-text for offline use. Built backend APIs and configured Alibaba Cloud services (DirectMail, OSS, Message Service) for production.

EDUCATION

APPLIED DATA SCIENCE & ARTIFICIAL INTELLIGENCE

Breda University of Applied Sciences

2023–2027

◇ Expected graduation: July 2027

◇ Focus: Data Engineering, Deep Learning, Computer Vision.

SELECTED PROJECTS

PREDICTIVE TRUCK MAINTENANCE SYSTEM *Industry Collaboration*

◇ Built a containerized anomaly detection pipeline for a heavy-vehicle fleet that integrates SQL, PDF telemetry, and RDW data to achieve 95.32% recall on failure predictions. By deploying an ensemble model, the system optimizes maintenance scheduling to target a 10–15% reduction in operational costs. Deliverables not publicly available (NDA)

TRANSPORT INCIDENT PREDICTION *ANWB Collaboration*

◇ Built machine learning models (XGBoost) to predict traffic incidents using weather patterns and historical data. Developed ETL pipelines with SQL, engineered features from messy real-world datasets, and optimized models to improve emergency response planning.

Deliverables not publicly available (NDA)

EMOTION CLASSIFICATION - DEPLOYMENT *End-to-End MLOps*

◇ Built a production-ready NLP pipeline with speech-to-text, preprocessing, and RoBERTa classification. Deployed via FastAPI and Docker on Azure ML with real-time inference. Demonstrates full lifecycle engineering from model training to cloud deployment.

[Link to the project](#)

NAC BREDA PLAYER VALUE ANALYSIS *Sports Analytics*

◇ Developed a data-driven framework for player value assessment for a professional football club. Utilized predictive modeling to estimate market value and analyzed performance metrics to support scouting decisions, adhering to strict GDPR standards.

[Link to the project](#)

AUTOMATED ROOT SEGMENTATION *Computer Vision*

◇ Developed a deep learning pipeline using the U-Net architecture to automate the segmentation and length measurement of plant roots. Implemented advanced image preprocessing with OpenCV and trained TensorFlow models to replace manual data collection, streamlining biological research.

[Link to the project](#)